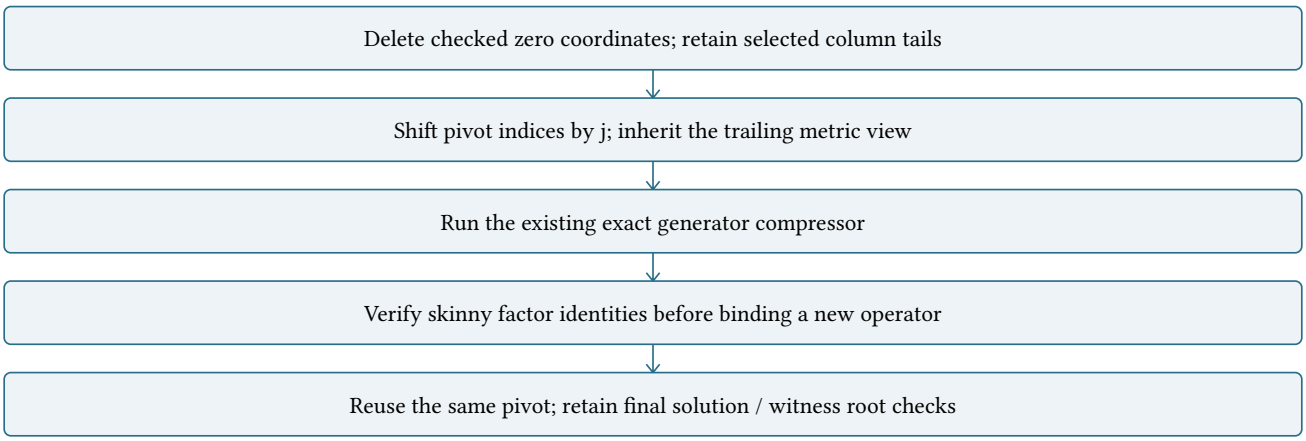
Restricted pivot reuse through compression

2026-11-02 · Derived schedule / static checks · Numerical trials: zero

Construction. Removing a proved-zero prefix shifts each selected index by j and restricts its cached column to the tail. The first column remains nonzero, so the reduced deterministic selector chooses the same scalar/pair pivot, including the secondary-diagonal branch.



Equality gate. The independent checker verifies the compressor's skinny factor identities and output generators. Equal displacement at fixed dimension and distinct companion wraps determines the same operator. The helper creates the rebound selection only after that check. The checker copy is byte-identical to the frozen source. This runtime gate has not yet been executed on instances.

Reduced selector branch	Source queries removed
First diagonal nonzero	1
Secondary diagonal or pair	2
Added compression-check products	2qg(s+t)
Added check modular reductions	q(s+2g+t)

Here q is restricted dimension, s input width, g first factor rank and t compressed width. The checker uses plain integer products and reductions; these derived schedules are separate from Meter field arithmetic. Each removed source query has 4t+8 instrumented native products in the preserved kernel, with dimension-dependent operand sizes. No net runtime gain is claimed.

Preservation and costs. Compression producer, pivot choice, paired formulas, Schur path, zero-padding frames, early witness rank semantics and root checks remain. Canonical cached-column tests, zero-prefix scans, metric-view matching, (k+1)q column copies and metric slices are charged as native work. The original search stays in the deletion trace; rebound search queries are zero.

Provenance. Source/meter/wrap/signature, dimensions, column shape, zero prefix, nonzero first tail and metric view are checked. A one-use receipt reserves rebinding before arithmetic. Compression equality is checked independently; deletion and original caches still require trusted internal provenance.

Validation. Two full static substitution audits cover the checker copy and complete driver integration; five Python files parse. Static inspection excludes helper source calls and places verification before operator/selection creation. Numerical, corruption, heap and performance trials remain zero. Funded tests must compare pivots/root outputs, corrupt certificate components, cover scalar tails and alternating frames, and independently check counters and lifetimes.

Sources / next. DERIVATION.md, reuse.py, driver_reuse.py, compression_certificate.py and integration.json. Bostan et al.2017 supplies the standard displacement conventions. Next consider sharing the checked compression boundary and measure prefix incidence when funding permits.